

**Emory University
Department of Data & Decision Sciences**

**QTM 340 (Fall 2026)
Text as Data**

Professor: Lauren Klein (lauren.klein@emory.edu)

Land Acknowledgment

Emory University is located on Muscogee (Creek) land. Emory was founded in 1836, during a period of sustained oppression, land dispossession, and forced removal of Muscogee (Creek) and Ani'yunwi'ya (Cherokee) peoples from Georgia and the Southeast. Emory owes an immense debt to the Muscogee, Ani'yunwi'ya and other original peoples, and their descendants, who have cared for and inhabited these lands.

Read the full [Land Acknowledgment and History Statement](#) developed by Emory faculty.

Dr. Klein's Office Hours:

Tuesdays 1-2pm and by appointment

Course Description

What does it mean to turn text into data? What are the data-scientific techniques that are commonly used to analyze text? How are they applied in the humanities and social sciences? How are they applied in the world? This course will explore these questions through a sustained semester-long project: training a large language model from scratch. Along the way, we will cover additional methods of text analysis including word and phrase similarity, text classification, and text clustering, as well as methods for creating, cleaning, and processing textual datasets. We will also discuss the economic and environmental issues, as well as the human costs, involved in our increasing reliance on large language models—including but not limited to the model we will create for this class.

Introductory courses in computer science and statistics are recommended (but not required) as prerequisites for this course. You will complete all programming exercises in Python, and all in-class assessments—including the final exam—in writing by hand. As part of your contribution to the class project, you will be expected to submit documentation in Markdown format, and create and give in-class presentations using slide software of your choosing. You will also be expected to take in-class notes in a notebook that I will provide.

Required Course Materials

All required readings will be posted on Canvas and/or are available online.

No-Screen Policy

Using personal laptops, phones, tablets, or other devices is not permitted in class, unless specified by the professor for use with a particular classroom assignment, such as a presentation or in-class coding exercise.

I have implemented this policy on the basis of significant research showing that use of cell phones, laptops, and other electronic devices cause distraction and negatively impact learning—not only for the person whose devices they are but also for the people around them. As you will see below, analog in-class notetaking has been incorporated into the design of the course.

Note: If you have accommodation needs around screens and/or note-taking that have been registered with the Department of Accessibility Services, I should have already received an email about this. As a next step, please schedule a meeting with me to discuss your specific needs.

List of Graded Assignments

Your grade for the course will be calculated as follows:

- Attendance and participation: 10%
- In-class notebook: 20%
- In class exams: 20% (10% each)
- Class project contribution (execution, deliverables, and presentation): 50% (25% each)

Description of Graded Assignments

Class Participation

Class participation is often assumed to be a hazy concept, but it actually involves a careful assessment in five distinct areas. Here are short descriptions of each of these areas, adapted from grading criteria developed by Dr. Mark Sample of Davidson College:

- **Preparation:** Reading/reviewing any assigned material before class.
- **Presence:** Being verbally and nonverbally engaged during class.
- **Focus:** Avoiding distractions during class (both in person and online).
- **Asking questions** in class and in office hours, as well as via email when appropriate.
- **Specificity:** Referring to specific ideas from readings and prior class discussions when contributing to class discussion and/or in conversations during office hours.

In-class notebook

As a way of increasing learning, maximizing retention, and enhancing class participation (see above), you will be required to take notes *by hand* in each class and, at times, respond to a formal in-class writing prompt. This work will take place in a notebook that I will provide to you, and will be evaluated three times over the course of the semester (see day-by-day schedule). After each evaluation cycle, you will receive a letter grade (A-F) as well as short written comments.

In-class exams

At the end of the first unit, and at the end of the semester, you will complete an in-class exam on the material covered in the course thus far. The exams are designed to test your *conceptual* understanding of the modeling approaches (and related techniques) that we will have discussed to that point, as well as your understanding of the ethical issues that they engage. You will be allowed to reference your notebook during the exam. The format will consist of multiple choice, short answer, and paragraph-long questions. We will spend time in class preparing for each exam. You will receive a letter grade (A-F) for each exam.

Class project contribution

The centerpiece of this course is our semester-long class project: training a large (ish) language model from scratch, using a public domain dataset of nineteenth-century texts from the British Library. This project is based on the [“Mr. Chatterbox” project](#) by Trip Venturella, who will be speaking to our class before we begin. The project has twelve steps, and we will work in two large groups in order to complete the steps in sequence along parallel tracks.

Each group will have fifteen students and each step will have two or three students assigned. At the beginning of the project unit, I will provide an overall plan and an outline of each step. The students assigned to each step will be required to take the step outline and implement it before the next class meeting; ***chatbots and coding assistants are allowed***. In the next class meeting, the students will present to the class about their process for implementing the step, and must be prepared to respond to questions about their decision-making process from me and from their classmates. They must also submit the following as a team:

- A copy of the presentation
- A formal log of the step
- A summary of the step’s cost and energy use (and water use if possible – tbd)
- A prompt to pass off to the students who will be taking over for the next step
- Any required code artifacts (if not stored on the cluster)

Individually, each student will submit a self-assessment of their personal contributions to the step and what they learned from the process. Each student will receive an ***individual*** letter grade (A-F) on the basis of their contribution to the step and the knowledge that they demonstrate during the presentation.

More detailed information about the project will be distributed as its start date approaches.

Policy on Late/Skipped Assignments

All assignments are mandatory. Should your group fail to complete its step before the next class, not only will you delay the entire project, but your grade for that assignment will decrease by a 1/3rd letter grade for each day that it is late. The only exception to this project is if your group

contacts ***me more than 24 hours*** before the step is due. Should you miss a notebook check or in-class exam, you will receive an F. The only exception is if you contact me ***more than 24 hours in advance*** to explain the situation. Should you experience difficulties working within your project group, please also contact me as soon as possible.

Grading Rubric

I will circulate a grading rubric for the class project contributions at the start of the first project unit in late September.

Process for Calculating Final Grades

At the end of the semester, I will convert each of your letter grades to a 12 point GPA scale (e.g. A = 12, A- = 11, B+ = 10) and weight each of these numbers according to the percentage listed above. On Canvas, the letter grade—NOT the numerical/percentage grade—will reflect your grade in the course.

Policy on Attendance and Punctuality

You are allowed two excused absences, no questions asked, with your grade beginning to be lowered with the third absence. With that said, ***I do not want to pressure you to come to class if you might be sick.*** If you miss class, you are responsible for making arrangements with a classmate to find out what was discussed in class, as I am not always available to meet before the next class meeting. I do not provide copies of my lecture notes, although I may post slide decks to Canvas. In addition, ***beginning with the third absence, you must email me to let me know that you will be missing class for health reasons.*** Finally, please be respectful to your peers and arrive on time. If you arrive more than 15 minutes late, you will be considered absent for that class.

Contacting Your Professor

I can be reached via my Emory email address. I respond to email M-F 9am-5pm, and outside of those hours if my schedule allows. Please allow 24 hours for a response, and 48 hours if your message is sent over the weekend.

Office of Accessibility Services

Office of Accessibility Services works with students who have disabilities to provide reasonable accommodations. In order to receive consideration for reasonable accommodations, you must contact OAS. It is the responsibility of the student to register with OAS. Please note that accommodations are not retroactive and that disability accommodations are not provided until an accommodation letter has been processed. Students registered with OAS who have a letter outlining their academic accommodations, are strongly encouraged to coordinate a meeting time with your professor that will be best for both to discuss a protocol to implement the accommodations as needed throughout the semester. This meeting should occur as early in the

semester as possible. Students must renew their accommodation letter every semester they attend classes. Contact the Office of Accessibility Services for more information at (404) 727-9877 or accessibility@emory.edu. Additional information is available at the OAS website at <http://equityandinclusion.emory.edu/access/students/index.html>.

Writing Center and ESL Program

Tutors in the Emory Writing Center and the ESL Program are available to support Emory College students as they work on any type of writing assignment, at any stage of the composing process. Tutors can assist with a range of projects, from traditional papers and presentations to websites and other multimedia projects. Writing Center and ESL tutors take a similar approach as they work with students on concerns including idea development, structure, use of sources, grammar, and word choice. They do not proofread for students. Instead, they discuss strategies and resources students can use as they write, revise, and edit their own work. Students who are non-native speakers of English are welcome to visit either the Writing Center tutors or the ESL tutors. All other students in the college should see Writing Center tutors. Learn more and make an appointment by visiting the websites of the ESL Program and the Writing Center. Please review tutoring policies before your visit.

Honor Code

The Honor Code applies to all work submitted for courses in Emory College. Students who violate the Honor Code may be subject to a written mark on their record, failure of the course, suspension, permanent exclusion, or a combination of these and other sanctions. The Honor Code may be reviewed online at: <http://catalog.college.emory.edu/academic/policies-regulations/honor-code.html>.

If you are unsure as to what constitutes plagiarism, please contact me before submitting your assignment.

Policy on Generative AI Tools in the Classroom

At this point, I assume that you are all at least somewhat familiar with the uses and limits of Generative AI (GenAI) tools (e.g. Claude, ChatGPT, Gemini, Copilot). In fact, one of the primary goals of this class is to allow you to become more knowledgeable about how these tools are created—and, therefore, about their uses and limits, and range of broader impacts. With that said, the landscape of GenAI in the classroom is confusing! For this reason, I have attempted to make clear guidelines about when GenAI tools can be used and not used in this class.

As such, **I am allowing the use of GenAI tools for all project-related programming assignments.** These assignments will almost certainly require specialized knowledge that I cannot cover in our class sessions.

During class it is a different story, however. Because in-class programming exercises are designed to teach you actual implementation skills, as well as underlying modeling and text analysis concepts, **the use of GenAI tools *other than those embedded into the Google Colab coding environment* for in-class programming exercises is prohibited.** This is because coding is a way of thinking, and coding is a way of learning. Very often, we do not entirely understand what is that we are need to do until we attempt to write actual code.

GenAI tools are also prohibited for all writing assignments, both in class and out of class. Like coding, writing is a way of thinking, and writing is a way of learning. I care very deeply about how you think, and how much you learn. I would much rather see less “perfect” writing that expresses your own original thinking than literally anything that genAI tools can produce.

While I cannot (and do not want to) police your use of GenAI tools for the required presentations, **I would strongly encourage you not to use GenAI to produce your slides for this class.** This is because you will be asked to defend your range of design choices—technical and otherwise—orally in front of the class, and if you have not put the time and thought into framing those choices, you will find it much more difficult to explain them.

I will require a formal genAI disclosure statement with all graded assignments in this course, and I will discuss this with you as the first deadline approaches.

Course Schedule

Lecture	Topic	Date	Milestone
First Week			
1	Course overview and syllabus	Weds Aug 26	
Unit 1: From Counting Words to Large Language Models			
2	Counting words <i>Sk-learn, logs odds, and a UFO dataset!</i>	Mon Aug 31	
3	Word vectors <i>Word2vec and a first look at neural networks</i>	Weds Sep 02	
—	<i>No class – Labor Day</i>	<i>Mon Sep 07</i>	
4	On the way to transformers <i>RNNs, LSTMs, contextual embeddings if time</i>	Weds Sep 09	
5	Intro to transformers <i>Attention, the transformer</i>	Mon Sep 14	
6	Unit 1 Review Day	Weds Sep 16	
7	In-class exam	Mon Sep 21	FIRST EXAM Due: Notebook check (everyone!)
Unit 2: Our Own Large Medium-Sized Language Model			
8	Project overview <i>Guest lecture: Trip Venturella (Mr. Chatterbox)</i>	Weds Sep 23	
9	Step 1: Learning about our cluster <i>Cloud infrastructure, data centers (incl. ours in Virginia), and the command line</i>	Mon Sep 28	
10	Step 2: Downloading / filtering our training data <i>pandas, regex, why libraries (of books) matter</i>	Weds Sep 30	Due: Step 1 (everyone!)
11	Step 3: Training the tokenizer <i>More on training data, tokenizers</i>	Mon Oct 05	Due: Step 2
12	Step 4: Pretraining the model	Weds	Due: Step 3

Lecture	Topic	Date	Milestone
	<i>More on attention, transformers</i>	Oct 07	
—	<i>No class – Fall Break</i>	<i>Mon Oct 12</i>	
Unit 3: Model Fine-Tuning			
13	Step 5: Dialogue extraction <i>spaCy, more regex, why domain expertise matters</i>	Weds Oct 14	Due: Step 4
14	Step 6: Quality filtering <i>Sentence embeddings, cosine similarity, sampling</i>	Mon Oct 19	Due: Step 5
15	Step 7: Question re-writing <i>Pretrained models, prompting, APIs</i>	Weds Oct 21	Due: Step 6
16	Step 8: Synthetic conversation generation <i>More prompting, clustering, classification</i>	Mon Oct 26	Due: Step 7 Due: notebook check (everyone!)
—	<i>Catch-Up Day – no class meeting (professor at Parsons)</i>	<i>Weds Oct 28</i>	
17	Step 9: Data evaluation, deduplication <i>Annotation, lexicons, a return to regex (always!)</i>	Mon Nov 02	Due: Step 8
18	Step 10: Supervised fine-tuning and evaluation, day 1 – behavioral <i>Fine-tuning and evaluation measures, human vs. machine evaluation</i>	Weds Nov 04	Due: Step 9
19	Step 11: SFT and evaluation, day 2 – instructions <i>Instruct vs. base models, creating an instruction dataset</i>	Mon Nov 09	Due: Step 10
—	<i>Catch-Up Day – no class meeting (professor at MIT)</i>	<i>Weds Nov 11</i>	
Unit 4: Deployment			
20	Step 12: Huggingface deployment <i>Huggingface, Gradio, model cards, the open source / open weight debate</i>	Mon Nov 16	Due: Step 11
21	Step 13: Celebrate! We trained a large language model!	Weds Nov 18	Due: Step 12

Lecture	Topic	Date	Milestone
—	<i>No class – Thanksgiving</i>	<i>Mon Nov 23</i>	
—	<i>No class – Thanksgiving</i>	<i>Weds Nov 25</i>	
Unit 5: Speculations / Course Wrap-Up			
22	RLHF (DPO too) <i>What is RLHF, who does RLHF, and at what cost?</i>	Mon Nov 30	
23	Benchmarks <i>What are benchmarks, why are they used, and what are their limits?</i>	Weds Dec 02	Due: Last notebook check
24	Catch-Up Day	Mon Dec 07	
25	Exam review	Weds Dec 09	
X	Final exam	Mon Dec 14	FINAL EXAM